March 31, 2026

Statement of Work

For

High Speed Spindle System

I. ***Background:***

The Air Force Institute of Technology needs a high speed spindle to perform engine relevant test conditions for modified turbine components. The proposed spindle would allow for engine matching conditions such as speed, thrust, load, heat flux, lubrication and air bypass.

II. ***Minimum Technical Requirements:***

The spindle drive system should consist of the following components:

1. 4 High Speed Spindle motors
2. 2 High frequency speed control drives
3. 2 Lubrication systems for spindle motor
4. 2 Cooling systems (if applicable)
5. 2 Full kits of additional cables, hoses, fittings, connectors, tools etc. to comprise a fully operational system
6. Assembly/Operation manuals

The minimum requirements for the spindle drive system to be delivered include the following:

High Speed Spindle Motor

1- Spindle RPM: Fully variable 15,000-90,000
2- Spindle Power: 3.5kW minimum at rated speed
3- Spindle interface: HJND-50 or smooth 14mm diameter shaft
4- Spindle feedback: Sensor/encoder for closed loop control
5- Spindle duty cycle: Continuous
6- Spindle Outside Diameter: 150mm (to interface with existing test infrastructure)
7- Spindle stand-alone refurbish/replacement [redacted]

High Frequency Speed Control Drive

1. Drive RPM: Fully variable 15,000-90,000
2. Drive control method: Closed Loop Speed Control
3. Drive power: 240/480V Single/Three phase 60Hz
4. Drive duty cycle: continuous
5. Drive communication interface: Discrete Digital/Analog I/O for enable and speed command and current/fault outputs

Lubrication System

1. Lube system method: Air/oil mist
2. Lube System Power: 240/480V Single/Three phase 60Hz
3. Lube System Air supply: 150psi max
4. Lube system air/oil filtration: included
5. Lube system communication interface: Discrete I/O to signal ready to run/fault condition to disable operation while lube system is in fault condition

Cooling system (if applicable)

1. Performance: Suitable for continuous spindle operation
2. Cooling system power: 240/480V Single/Three phase 60Hz
3. Cooling system communication interface: Discrete I/O to signal ready to run/fault condition to disable operation while cooling system is in fault condition

III. ***Training:*** No training required

IV. ***Installation:*** No installation required

V. ***Delivery:***

SHIP TO DoDAAC: F4F5AV
Air Force Institute of Technology
AFIT/ENY
2950 Hobson Way
WPAFB, OH 45433
POC: Matthew Brandham, matthew.branham.2@us.af.mil, (385) 633-3245,

VI. ***Lead Time:***

The contractor shall deliver ordered systems no later than 26 weeks after contract approval.

VII. ***Warranty:*** 1 year after receipt of delivery

VIII. ***Delivery Procedures – Commercial Vehicles:***

a. The company shall properly ship the system to the Air Force Institute of Technology at WPAFB, OH. All shipping costs to include insurance will be included in contractor's proposal. Great care using standard, approved packaging methods shall be taken.

b. All vehicles larger than a large pick-up truck are required to be inspected by the Wright-Patterson Air Force Base Commercial Vehicle Delivery Gate (CVDG) prior to entering the installation. Vehicles to be inspected include, but are not limited to, the following:
   1. Step van/panel truck
   2. Tractor/trailer, box and flatbed containing cargo
   3. Tanker trucks
   4. Box trucks
   5. Tour buses
   6. Garbage/recycled waste trucks
   7. Concrete trucks/mixers, dump trucks
   8. Cranes, recreational vehicles, petroleum tanker

This inspection will be conducted at Gate 26A located off State Route 235.

c. The following are exemptions to vehicles utilizing the CVDG:

   1. If the vehicle has the product inside (concrete and asphalt trucks) and timely delivery is necessary due to product deterioration it does not need to enter the CVDG. To bypass the CVDG, the contractor shall submit a list containing drivers' names, social security numbers and the state in which the driver's license is held for those drivers who will be entering the base. This shall be accomplished 24 hours prior to requested entry time. If entry is requested on Monday, this list must be submitted by Friday at 1630 hours. All lists shall be submitted to the 88th ABW/CE Directorate contract inspector. The only gates that may be used under this exemption shall be 15A, 26A, 38A, and gate 1B. If the driver's name is not on the list, he/she will not be allowed access to the installation through these gates and the base will not assume liability for denied access.

   2. If a delivery vehicle must exit, and then re-enter the base to complete its route, the vehicle shall be resealed upon exiting the base. After initially passing through the commercial vehicle delivery gate, trucks shall be resealed at Gates 15A, 38A and 22B. The resealing of the trucks will allow them to continue to any other area of the installation (Areas A, B, or Kittyhawk) without reprocessing through the CVDG. To receive resealing assistance, the drivers shall physically stop at one of the three authorized gates and request the installation entry controller to reseal their truck and provide the next location of their delivery. The controller will reseal the truck and give the delivery driver a pre-clearance form. The driver shall present the pre-clearance form to the entry controller at the next point of installation entry. This reentry can be through any base gate.

d. Vehicles may be subject to an inspection at any of installation entry control points during a directed random antiterrorism measure (RAM.) Any commercial vehicle, regardless of size, can be directed to the CVDG at the discretion of the installation entry controller.

VIV. ***Invoice and payment.***

a. Invoice and Payment: 100% payment after delivery. The contractor shall be paid upon submission of a properly prepared invoice and after final inspection and acceptance have been made. The invoice shall be submitted to in Wide Area Workflow (WAWF) in accordance with the terms of the final contract award.